KUNLUN LI

APPLICATION ALGORITHM ENGINEER | MEDICAL IMAGING & MACHINE LEARNING

Beijing, China | lkl2654462334@outlook.com
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PROFESSIONAL EXPERIENCE

Application Algorithm Engineer | blueballon Mar 2025 - Present | Beijing, China

RESEARCH EXPERIENCE

Graduate Research Assistant | Johns Hopkins University May 2024 - Jan 2025 | Baltimore, MD

Worked with Prof. Eric Nalisnick on depth-enhanced sign language recognition using monocular depth estimation, semi-supervised learning, and active learning under limited annotation.

Summer Research Program | UCLA Jun 2021 - Sep 2021 | Los Angeles, CA

Studied gate-sizing optimization for digital integrated circuits, balancing delay and energy consumption under the supervision of Prof. Dejan Markovic.

EDUCATION

M.S., Electrical and Computer Engineering | Johns Hopkins University Aug 2023 - Jan 2025

B.E., Electrical Engineering | University of Mississippi Aug 2021 - May 2023

B.E., Electrical Information Engineering | North China University of Technology Sep 2019 - Jun 2021

SELECTED RESEARCH PROJECTS

Deep Learning-Based Liver and Tumor Segmentation 2024

Compared U-Net, Attention U-Net, and DALU-Net on 3D-IRCADb CT data. The project report records a DALU-Net Dice coefficient of 0.899, compared with 0.822 for the U-Net baseline.

Evaluation of Subarachnoid-Space Volume in MRI 2024

Built a coursework pipeline using FSL, FastSurfer, CycleGAN, K-means clustering, and morphology for MRI registration, synthesis, segmentation, and quantitative analysis.

Epoch Extraction for Speaker Differentiation 2023 - 2024

Implemented zero-frequency filtering and a locking-based comparison workflow for CMU ARCTIC speech data, with an implementation path toward Raspberry Pi and hardware-oriented systems.

SELECTED PUBLICATIONS AND MANUSCRIPTS

- **K. Li**, D. Ferro, X. Zhao, A. J. Syed, A. K. Vuppala, and A. Syed. "Matlab-based Epoch Extraction for Speaker Differentiation." *arXiv:2407.18447*, 2024. arXiv
- **K. Li**, J. Ma, and Y. Zhang. "Optimization of a Three-Bit Absolute Value Detector with Domino Logic and Gate Sizing." *Proceedings of SPIE 12161*, 2022. doi:10.1117/12.2627237
- R. Chen, **K. Li**, Y. Wang, and E. Nalisnick. "SSLAL: Leveraging Semi-Supervised and Active Learning for Robust Depth-Enhanced Sign Language Recognition." Unpublished manuscript, 2024.

TECHNICAL SKILLS

Programming: Python, MATLAB | **Frameworks and tools:** PyTorch, FSL, FastSurfer, MIPAV, LaTeX | **Languages:** Mandarin (native), English (fluent), Japanese (beginner)